

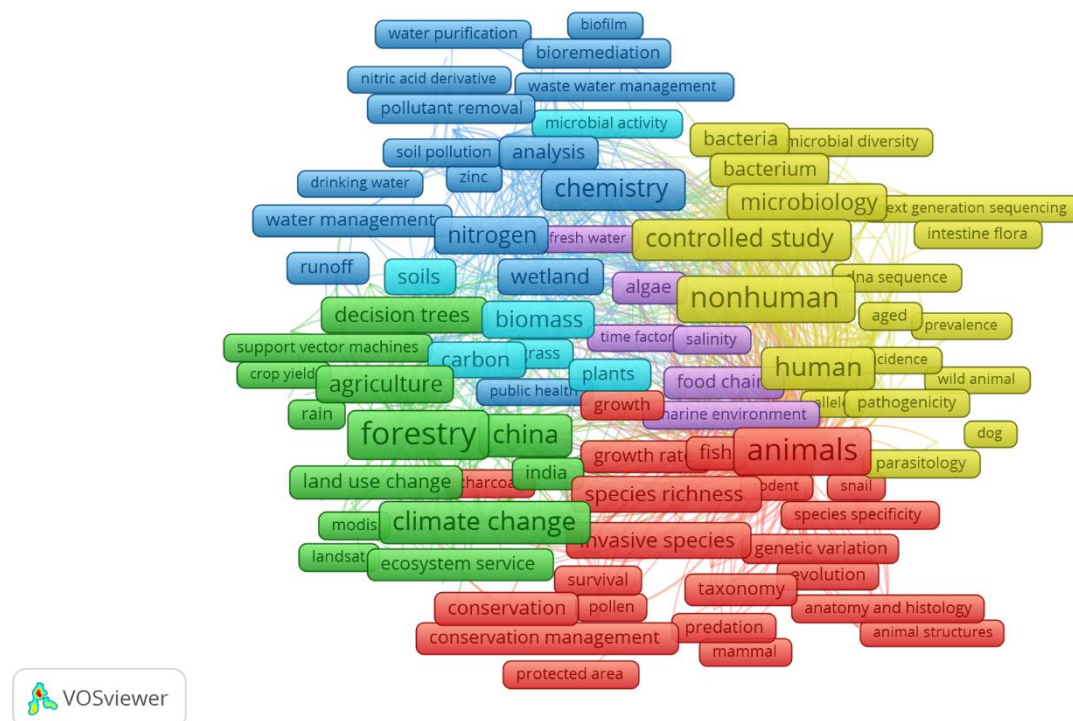
SDG15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

SUMMARY

Original search string: 409,711 document results

TITLE-ABS-KEY((biodivers* OR bio-divers* OR bioeconom* OR bio-econom* OR {biological production} OR deforest* OR desertif* OR {earth system} OR {ecological resilience} OR ecosystem* OR eco-system* OR {food chain} OR {food chains} OR {threatened species} OR {endangered species} OR {extinction risk} OR {extinction risks} OR poach* OR {wildlife product} OR {wildlife products} OR {wildlife traffic} OR {wildlife market} OR {wildlife markets} OR {wildlife trafficking} OR {invasive species} OR {alien species} OR {land uses} OR {land use} OR {land uses} OR {land degradation} OR {soil degradation} OR {LULUCF} OR *forest* OR {land conservation} OR wetland* OR mountain* OR dryland*)) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



Research string after second (and last) update: 97,814 document results

TITLE-ABS-

KEY ((terrestrial OR land OR inland OR freshwater) AND (biodivers* OR {species richness} OR bioeconom* OR bio-econom* OR {biological production} OR deforest* OR desertif* OR {earth system} OR {ecological resilience} OR ecosystem* OR eco-system* OR {trophic cascade} OR {trophic level} OR {trophic web} OR {threatened species} OR {endangered species} OR {extinction risk} OR {extinction risks} OR poach* OR {wildlife product} OR {wildlife products} OR {wildlife traffic} OR {wildlife market} OR {wildlife markets} OR {wildlife trafficking} OR {invasive species} OR {alien species} OR {land uses} OR {land use} OR {land uses} OR {land degradation} OR {soil degradation} OR {LULUCF} OR *forest* OR {land conservation} OR wetland* OR mountain* OR dryland* OR {mountainous cover} OR {protected area} OR {protected areas} OR {REDD} OR {forest management} OR {silviculture} OR {timber harvest} OR {illegal logging} OR {slash-and-burn} OR {fire-fallow cultivation} OR {tree cover} OR {soil restoration} OR {land restoration} OR {drought} OR {sustainable land management} OR {mountain vegetation} OR {habitat restoration} OR {Red List species} OR {Red List Index} OR {extinction wave} OR {habitat fragmentation} OR {habitat loss} OR {Nagoya Protocol on Access to Genetic Resources} OR {genetic resources} OR {biological invasion} OR {biodiversity-inclusive} OR {forest stewardship council} OR {rainforest alliance} OR {forest certification} OR {forest auditing} OR {ecotourism} OR {community-based conservation} OR {community based conservation} OR {human-wildlife conflict})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

PROCESS

The first stage of this review was to compare the original search string to the targets listed under the Sustainable Development Goal (SDG).

[SDG targets listed here: https://sustainabledevelopment.un.org/sdg15](https://sustainabledevelopment.un.org/sdg15)

The purpose of this stage was to understand whether the Scopus results correctly reflected the theme of the SDG. The SDGs were unanimously agreed by member states of the United Nations, with very deliberate language and scope/ exclusions, so the results of this analysis should accurately reflect the specific SDG themes.

The SDG targets show that the overall theme aims to encourage the **protection, restoration and sustainable use terrestrial and inland freshwater ecosystems and biodiversity**. **Degradation of natural resources and habitats** is aimed to be stopped, **sustainable management practices** and restoration activities such as afforestation are promoted. There is a specific emphasis on **forests, desertification, soil degradation and mountainous ecosystems**. Regarding biodiversity the SDG aims to halt the **decline of biodiversity**, to end **wildlife poaching** and **trafficking** and to prevent the spread of **invasive species**.

A Scopus search was conducted using the original search string. This returned 409,711 document results.

The top 20 cited articles were reviewed. 13 of the 20 articles did not closely reflect the theme of the SDG. Specifically, articles generally related to microbiology (DNA / RNA), medicine, chemistry or IT (with hits on ambiguous keywords such as ecosystem, forest, tree).

Top 20 most cited articles listed in Scopus from the original search string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
The SILVA ribosomal RNA gene database project: Improved data processing and web-based tools	2013	Nucleic Acids Research	3921	2-s2.0.8487373931
Cancer genome landscapes	2013	Science	3263	2-s2.0.8487549018
High-resolution global maps of 21st-century forest cover change	2013	Science	2810	2-s2.0.8488772446
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1609	2-s2.0.8492243339
Köppen's climate classification map for Brazil (?)	2013	Meteorologische Zeitschrift	1513	2-s2.0.8489710040
Graphitic Carbon Nitride (g-C ₃ N ₄)-Based Photocatalysts for Artificial Photosynthesis and Environmental Remediation: Are We a Step Closer to Achieving Sustainability?	2016	Chemical Reviews	1492	2-s2.0.8496644769
Evaluation of general 16S ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies	2013	Nucleic Acids Research	1476	2-s2.0.8487180849
Recursive deep models for semantic compositionality over a sentiment treebank	2013	EMNLP 2013 - 2013 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Conference	1469	2-s2.0.8492635884

Macrophage biology in development, homeostasis and disease	2013	Nature	1377	2-s2.0-8487680033
XGBoost: A scalable tree boosting system	2016	Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining	1297	2-s2.0-8498495069
Changes in the global value of ecosystem services	2014	Global Environmental Change	1234	2-s2.0-8490140697
Long-term climate change: Projections, commitments and irreversibility	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1191	2-s2.0-8492860076
Single-cell RNA-seq highlights intratumoral heterogeneity in primary glioblastoma	2014	Science	1032	2-s2.0-8490266880
Completion of the 2011 national land cover database for the conterminous United States – Representing a decade of land cover change information	2015	Photogrammetric Engineering and Remote Sensing	1026	2-s2.0-8492946556
Impacts of biological invasions: What's what and the way forward	2013	Trends in Ecology and Evolution	1013	2-s2.0-8487169685
Natural products: A continuing source of novel drug leads	2013	Biochimica et Biophysica Acta - General Subjects	997	2-s2.0-8487600267
Enhanced nitrogen deposition over China	2013	Nature	990	2-s2.0-8487436900
Phytoremediation of heavy metals-Concepts and applications	2013	Chemosphere	961	2-s2.0-8488437695
New handbook for standardised measurement of plant functional traits worldwide	2013	Australian Journal of Botany	961	2-s2.0-8487809379
XSEDE: Accelerating scientific discovery	2014	Computing in Science and Engineering	939	2-s2.0-8490809918

2-s2.0-85033782500

In addition, a randomised sample of 20 articles was reviewed. A randomised sample was drawn from 200,000 document results that were retrieved by using the original search query. Eight of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles generally related to microbiology, chemistry and IT. Additionally, non-scientific articles such as opinions and fiction (sages) were classified as 'not closely related'. Hits on keywords such as 'wood' or 'mountain' led to the retrieval of these articles. The topics of these articles relate to nature or a natural habitat, but these articles do not provide scientific data related to the SDG. Therefore, it has been decided to classify these as 'not closely related'.

Randomised sample of 20 articles listed in Scopus from the original search string:

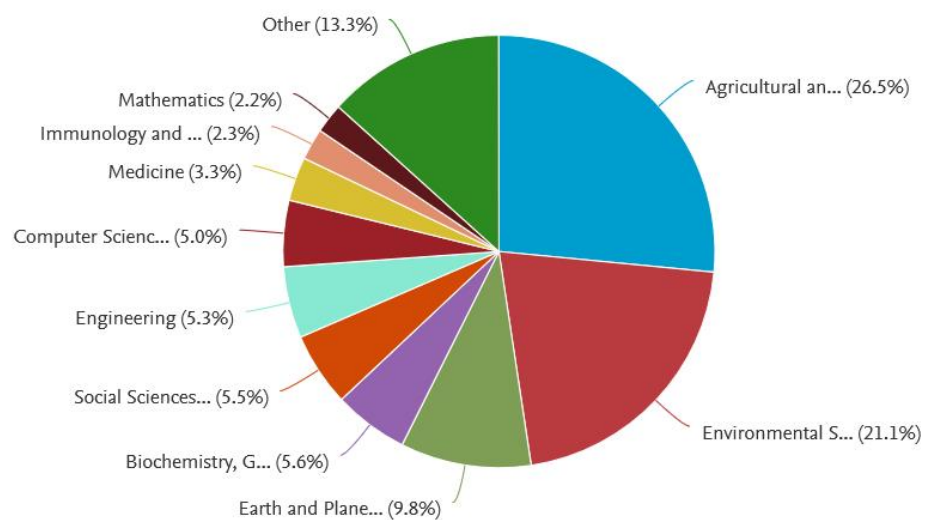
*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
The size-distribution of Earth's lakes	2016	Scientific Reports	23	2-s2.0-8497812115
Impact Evaluation of Forest Conservation Programs: Benefit-Cost Analysis, Without the Economics	2016	Environmental and Resource Economics	16	2-s2.0-8495784110
Near-Channel Versus Watershed Controls on Sediment Rating Curves	2017	Journal of Geophysical Research: Earth Surface	12	2-s2.0-8503378250
An Approximation Algorithm for the Maximum-Lifetime Data Aggregation Tree Problem in Wireless Sensor Networks	2017	IEEE Transactions on Wireless Communications	12	2-s2.0-8502093486
Evaluating mountain meadow groundwater response to Pinyon-Juniper and temperature in a great basin watershed	2017	Ecohydrology	11	2-s2.0-8500598099
The impact of charcoal and soil mixtures on decomposition and soil microbial communities in boreal forest	2016	Applied Soil Ecology	7	2-s2.0-8494840601
Lidar-derived variables as a proxy for fungal species richness and composition in temperate Northern Europe	2017	Remote Sensing of Environment	5	2-s2.0-8502892623
Preservation methods alter carbon and nitrogen stable isotope values in crickets (orthoptera: Grylloidea)	2015	PLoS ONE	4	2-s2.0-8494655414
The Consequences of Nonrandomness in Species-Sensitivity in Relation to Functional Traits for Ecosystem-Level Effects of Chemicals	2017	Environmental Science and Technology	3	2-s2.0-8502162914
Reproductive investment of Pinus pseudostrobus along an altitudinal gradient in Western Mexico: implications of climate change	2017	New Forests	2	2-s2.0-8502182069
Evidence of microbial rhodopsins in Antarctic Dry Valley edaphic systems	2017	Environmental Microbiology	1	2-s2.0-8502795905
Soil fertility evaluation of artificial afforestation areas in Tibet, China	2017	Chinese Journal of Applied Ecology	1	2-s2.0-8502654707
Comparison of fine-scale spatial genetic structure of two sympatric Rhododendron shrub species in forest habitat having different seed weights: A case study	2016	Silvae Genetica	1	2-s2.0-8505371199
One-step hydrothermal synthesis and characterization of the ZnO/hydrothermal carbon composite material	2017	Beijing Linye Daxue Xuebao/Journal of Beijing Forestry University	0	2-s2.0-8502980782
Diversity of the auxotrophic requirements in natural isolates of Escherichia coli	2017	Microbiology (United Kingdom)	0	2-s2.0-8502219323
Total organic matter profile in shrimp-seaweeds polyculture system	2017	Advanced Science Letters	0	2-s2.0-8501859050

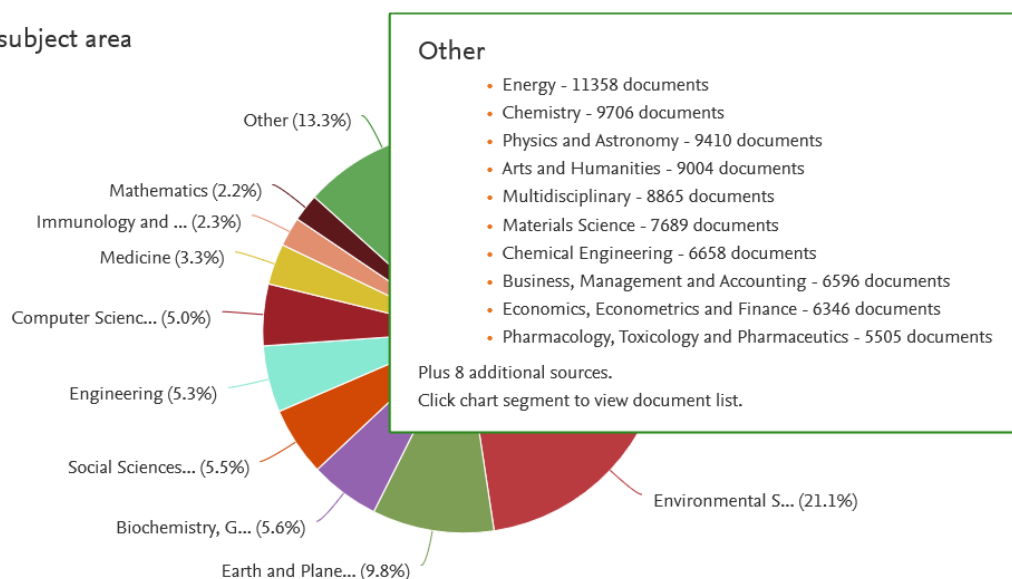
Assessing and understanding the environmental changes in the Doubegue watershed (Burkina Faso) by combining temporal study of land use changes and stakeholders' knowledge	2017	Geographical Journal	0	2-s2.0-8495667396
Natural history: Voices from the greenwood	2016	Nature	0	2-s2.0-8499237482
The old of the Mountain, or the virtues of the counter-example: The Seven Sages of Rome, Barlaam and Josaphat, the Decameron [Le Vieux de la Montagne, ou les vertus du contre-exemple: Le Roman des Sept Sages, Barlaam et Josaphat, Le D Crossed D sign©camCrossed D sign©ron]	2016	French Studies	0	2-s2.0-8499744775
Building ecosystem-aware tools using the ecosystem monitoring framework	2016	CEUR Workshop Proceedings	0	2-s2.0-8501610735

Reflection on subject areas that should be excluded (incl. pie charts)

Documents by subject area



Documents by subject area



The general analysis of the total document results confirms the results found in the different analyses. The fields 'Agricultural and Biological Sciences' and 'Environmental Sciences' cover nearly half of the document results. However, this does not mean these are all relevant for the SDG as these could also cover the deposit of specific pollutants, climate change or marine environments. Disregarding 'Other' and its subfields, 'Biochemistry, Genetics and Microbiology', 'Engineering' and 'Computer Sciences' all take up a significant proportion of the results, which matches the previous findings. Additionally, the large proportion of articles related to 'Energy sciences' and 'Chemistry' is reflected in the general analysis. These fields are not relevant for SDG15. This search query could be improved to be able to retrieve more relevant document results. It will be important to aim to exclude ambiguous keywords or associated fields that are not relevant for the SDG, such as IT. It will also be important to narrow down keywords to achieve more hits on relevant documents and to exclude closely related documents results on marine environments. Marine environments fall within the scope of SDG 14 and should therefore be excluded from SDG 15.

Update 1) The search keywords were edited to remove the broader search terms, or to make them more specific by combining with other search keywords.

It is crucial to rewrite this search query to deliver a query that specifically focusses on terrestrial or freshwater systems. It is expected that a large proportion of articles that are not closely related to the theme of the SDG (e.g. IT or microbiology studies) are removed by doing this. In addition, the following changes were made to the specific query terms:

Original search string: 409,711 document results

TITLE-ABS-KEY((biodivers* OR bio-divers* OR bioeconom* OR bio-econom* OR {biological production} OR deforest* OR desertif* OR {earth system} OR {ecological resilience} OR ecosystem* OR eco-system* OR {food chain} OR {food chains} OR {threatened species} OR {endangered species} OR {extinction risk} OR {extinction risks} OR poach* OR {wildlife product} OR {wildlife products} OR {wildlife traffic} OR {wildlife market} OR {wildlife markets} OR {wildlife trafficking} OR {invasive species} OR {alien species} OR {land uses} OR {land use} OR {land uses} OR {land degradation} OR {soil degradation} OR {LULUCF} OR *forest* OR {land conservation} OR wetland* OR mountain* OR dryland*)) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Rationale for edits to the search keywords:

query term	decision	Rationale
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Bio-divers*	Deleted	There is no keyword such as bio-diversity used in biodiversity studies, deleted to clean up the query
{species richness}	Added	In addition to biodiverse*
{food chain}	Changed to {trophic cascade} OR {trophic level} OR {trophic web}	Not necessarily a target of this SDG, and will lead to hits on food security. Narrowed down to include articles on trophic cascade / trophic levels within a biodiversity food chain
{food chains}	Deleted	See {food chain}

The query has been rewritten by adding the following query terms to merely include hits related to terrestrial or freshwater systems: (terrestrial OR land OR inland OR freshwater) AND ...

Updated search string (first update): 94,819 document results

TITLE-ABS-

KEY ((*terrestrial* OR *land* OR *inland* OR *freshwater*) AND (*biodivers** OR {*species richness*} OR *bioeconom** OR *bio-econom** OR {*biological production*} OR *deforest** OR *desertif** OR {*earth system*} OR {*ecological resilience*} OR *ecosystem** OR *eco-system** OR {*trophic cascade*} OR {*trophic level*} OR {*trophic web*} OR {*threatened species*} OR {*endangered species*} OR {*extinction risk*} OR {*extinction risks*} OR *poach** OR {*wildlife product*} OR {*wildlife products*} OR {*wildlife traffic*} OR {*wildlife market*} OR {*wildlife markets*} OR {*wildlife trafficking*} OR {*invasive species*} OR {*alien species*} OR {*land uses*} OR {*land use*} OR {*land uses*} OR {*land degradation*} OR {*soil degradation*} OR {*LULUCF*} OR **forest** OR {*land conservation*} OR *wetland** OR *mountain** OR *dryland**)) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 94,819 document results, compared to 409,711 document results for the original search. The top 20 cited articles were reviewed. Six of the 20 articles do not closely reflect the theme of the SDG. These articles covered topics related to climate change, human geography, medicine or marine systems. Hits were found on the keywords 'land use', 'forest' and 'terrestrial ecosystem'.

Top 20 most cited articles listed in Scopus from search string version 1:

**red text shows articles which do not closely match the theme of the SDG*

Title	Year	Source title	Cited by	EID
High-resolution global maps of 21st-century forest cover change	2013	Science	2870	2-s2.0-84887724460
Changes in the global value of ecosystem services	2014	Global Environmental Change	1257	2-s2.0-84901406973
Completion of the 2011 national land cover database for the conterminous United States – Representing a decade of land cover change information	2015	Photogrammetric Engineering and Remote Sensing	1050	2-s2.0-84929465562
Enhanced nitrogen deposition over China	2013	Nature	1001	2-s2.0-84874369007
Status and ecological effects of the world's largest carnivores	2014	Science	918	2-s2.0-84892185014

Defaunation in the Anthropocene	2014	Science	858	2-s2.0-84904795599
Communities and ecosystems: Linking the aboveground and belowground components	2013	Communities and Ecosystems: Linking the Aboveground and Belowground Components	797	2-s2.0-84890745455
The biodiversity of species and their rates of extinction, distribution, and protection	2014	Science	779	2-s2.0-84901642271
Comprehensive evaluation of antibiotics emission and fate in the river basins of China: Source analysis, multimedia modeling, and linkage to bacterial resistance	2015	Environmental Science and Technology	769	2-s2.0-84930632873
Climate change projections using the IPSL-CM5 Earth System Model: From CMIP3 to CMIP5	2013	Climate Dynamics	712	2-s2.0-84876880770
Policy: Sustainable development goals for people and planet	2013	Nature	680	2-s2.0-84875346108
Paris Agreement climate proposals need a boost to keep warming well below 2 °c	2016	Nature	678	2-s2.0-84977119288
Global effects of land use on local terrestrial biodiversity	2015	Nature	676	2-s2.0-84926360683
Climate change impacts on global food security	2013	Science	669	2-s2.0-84881061285
Urban green space, public health, and environmental justice: The challenge of making cities 'just green enough'	2014	Landscape and Urban Planning	644	2-s2.0-84899128460
Nitrous oxide emissions from soils: How well do we understand the processes and their controls?	2013	Philosophical Transactions of the Royal Society B: Biological Sciences	633	2-s2.0-84878241957
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	624	2-s2.0-84878224971
Climate change, phenology, and phenological control of vegetation feedbacks to the climate system	2013	Agricultural and Forest Meteorology	612	2-s2.0-84869873408
Global imprint of climate change on marine life	2013	Nature Climate Change	611	2-s2.0-84886722508
Improved allometric models to estimate the aboveground biomass of tropical trees	2014	Global Change Biology	593	2-s2.0-84908499511

In addition, a randomised sample of 20 articles was reviewed. Five of the 20 articles were not closely related to the theme of the SDG. These articles covered a variety of topics: marine systems, air pollution (unrelated to soil degradation), healthcare, recreation studies and spatial planning.

Randomised sample of 20 articles listed in Scopus from search string version 1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Soil methane oxidation and land-use change - from process to mitigation	2015	Soil Biology and Biochemistry	67	2-s2.0-84909957718

FORUM: Sharing or sparing? How should we grow the world's cities?	2013	Journal of Applied Ecology	62	2-s2.0-84890859631
Land-use history and topographic gradients as driving factors of subalpine <i>Larix decidua</i> forests	2013	Landscape Ecology	32	2-s2.0-84876985708
Macropredatory ichthyosaur from the Middle Triassic and the origin of modern trophic networks	2013	Proceedings of the National Academy of Sciences of the United States of America	30	2-s2.0-84872832850
Assessment of the magnitude and recent trends in satellite-derived ground-level nitrogen dioxide over North America	2015	Atmospheric Environment	21	2-s2.0-84940403929
Influence of cohesive and disruptive forces on strength and erodibility of tropical soils	2013	Soil and Tillage Research	16	2-s2.0-84879697773
Responses of nutrient concentrations and stoichiometry of senesced leaves in dominant plants to nitrogen addition and prescribed burning in a temperate steppe	2014	Ecological Engineering	11	2-s2.0-84901981218
The impact of ecotourism employment on rural household incomes and social welfare in six southern African countries	2014	Tourism and Hospitality Research	10	2-s2.0-84906665938
The impact of land use change on migrant birds in the Sahel	2014	Biodiversity	9	2-s2.0-84904272826
New record and phylogenetic affinities of the oomycete <i>Olpidopsis feldmanni</i> infecting <i>Asparagopsis</i> sp. (Rhodophyta)	2015	Diseases of Aquatic Organisms	8	2-s2.0-84947738799
Nitrogen and phosphorus fractions and releasing characteristics of the soils from the representative water-level-fluctuating zone of Danjiangkou Reservoir	2015	Huanjing Kexue Xuebao/Acta Scientiae Circumstantiae	6	2-s2.0-84930516839
Seasonal dynamics of the shoreline vegetation in the Zapotosa floodplain lake complex, Colombia [Dinámica estacional de la vegetación litoral en el complejo lagunar inundable Zapotosa, Colombia]	2014	Revista de Biología Tropical	4	2-s2.0-84918798668
Carbon sequestration in soil following afforestation of former agricultural land in the Czech Republic	2017	Central European Forestry Journal	3	2-s2.0-85038364766
Characteristics of automated external defibrillator coverage in Philadelphia, PA, based on land use and estimated risk	2016	Resuscitation	3	2-s2.0-84990204466
Immersing into another world - A study on concepts of recreation and recreation potential in the forest [Eintauchen in eine andere welt - Untersuchung iiber erhölungskonzepte und erhölungsprozesse im wald forstliche versuchs- und forschungsanstalt baden-württemberg]	2013	Allgemeine Forst- und Jagdzeitung	3	2-s2.0-84879208832
The Whitewood (<i>Endospermum Medullosum</i>) Value-Chain in Vanuatu and Impediments to Development of a Plantation-Based Industry	2015	Small-scale Forestry	2	2-s2.0-84930212791
More than just a Long Paddock: Fostering native vegetation recovery in Riverina Travelling Stock Routes and Reserves	2017	Ecological Management and Restoration	1	2-s2.0-85010375259
Inventorying Armagh: Max Lock, civil society, and the diffusion of planning ideas into Northern Ireland in the 1960s	2017	Planning Perspectives	0	2-s2.0-85012108896

Title	Year	Source title	Cited by	EID
The SILVA ribosomal RNA gene database project: Improved data processing and web-based tools	2013	Nucleic Acids Research	4064	2-s2.0-84873739311
Cancer genome landscapes	2013	Science	3314	2-s2.0-84875490185
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1651	2-s2.0-84922433390
Köppen's climate classification map for Brazil	2013	Meteorologische Zeitschrift	1559	2-s2.0-84897100402
Graphitic Carbon Nitride (g-C ₃ N ₄)-Based Photocatalysts for Artificial Photosynthesis and Environmental Remediation: Are We a Step Closer to Achieving Sustainability?	2016	Chemical Reviews	1531	2-s2.0-84966447697
Evaluation of general 16S ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies	2013	Nucleic Acids Research	1523	2-s2.0-84871808499
Recursive deep models for semantic compositionality over a sentiment treebank	2013	EMNLP 2013 - 2013 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Conference	1499	2-s2.0-84926358845
Macrophage biology in development, homeostasis and disease	2013	Nature	1394	2-s2.0-84876800337
XGBoost: A scalable tree boosting system	2016	Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining	1377	2-s2.0-84984950690
Long-term climate change: Projections, commitments and irreversibility	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1207	2-s2.0-84928600768
Single-cell RNA-seq highlights intratumoral heterogeneity in primary glioblastoma	2014	Science	1051	2-s2.0-84902668801
Impacts of biological invasions: What's what and the way forward	2013	Trends in Ecology and Evolution	1029	2-s2.0-84871696859
Natural products: A continuing source of novel drug leads	2013	Biochimica et Biophysica Acta - General Subjects	1010	2-s2.0-84876002671
New handbook for standardised measurement of plant functional traits worldwide	2013	Australian Journal of Botany	980	2-s2.0-84878093794
Phytoremediation of heavy metals-Concepts and applications	2013	Chemosphere	979	2-s2.0-84884376957
Quality-filtering vastly improves diversity estimates from Illumina amplicon sequencing	2013	Nature Methods	964	2-s2.0-84871979584

XSEDE: Accelerating scientific discovery	2014	Computing in Science and Engineering	957	2-s2.0-84908099185
The Culture of Connectivity: A Critical History of Social Media	2013	The Culture of Connectivity: A Critical History of Social Media	949	2-s2.0-84917742994
Insights into the phylogeny and coding potential of microbial dark matter	2013	Nature	890	2-s2.0-84881138595
Do we need hundreds of classifiers to solve real world classification problems?	2014	Journal of Machine Learning Research	867	2-s2.0-84919773193

The analyses show that the updated search query retrieves a higher number of relevant document results compared to the original search query. A large number of IT – and healthcare related articles are excluded from the updated search results. The co-occurrence of the keywords shows that there is a more specific focus on topics related to the theme of the SDG. Still, overlap is witnessed with climate change studies that mention land use in their keywords. Climate change models often use land use modelling as an input, but do not necessarily relate to SDG 15. Other articles related to the natural environment, however not with a focus on protection such as recreational studies or public health and greenspaces, show that further improvements are still required. In addition, some targets are not or inadequately covered by the current search string. Keywords will need to be added to ensure articles related to these targets are retrieved by the search query.

Update 2) A second update was made to the search string to cover topics such as ...

To bring these articles into the search results, the following were added to the search string:

Target	Keywords (to be added)
By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements	{mountainous cover} {protected area} {protected areas}
By 2020, promote the implementation of sustainable management of all types of forests , halt deforestation , restore degraded forests and substantially increase afforestation and reforestation globally	{REDD} {forest management} {silviculture} {timber harvest} {illegal logging} {slash-and-burn} {fire-fallow cultivation} {tree cover}
By 2030, combat desertification, restore degraded land and soil , including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world	{soil restoration} {land restoration} {drought} {sustainable land management}
By 2030, ensure the conservation of mountain ecosystems , including their biodiversity, in order to enhance their capacity to provide benefits that are essential for sustainable development	{mountain vegetation}
Take urgent and significant action to reduce the degradation of natural habitats, halt the loss	{habitat restoration} {Red List species}

of biodiversity and, by 2020, protect and prevent the extinction of threatened species	{Red List Index} {extinction wave} {habitat fragmentation} {habitat loss}
Promote fair and equitable sharing of the benefits arising from the utilization of genetic resources and promote appropriate access to such resources, as internationally agreed	{Nagoya Protocol on Access to Genetic Resources} {genetic resources}
Take urgent action to end poaching and trafficking of protected species of flora and fauna and address both demand and supply of illegal wildlife products	Included
By 2020, introduce measures to prevent the introduction and significantly reduce the impact of invasive alien species on land and water ecosystems and control or eradicate the priority species	{biological invasion}
By 2020, integrate ecosystem and biodiversity values into national and local planning , development processes, poverty reduction strategies and accounts	{biodiversity-inclusive}
Mobilize and significantly increase financial resources from all sources to conserve and sustainably use biodiversity and ecosystems	
Mobilize significant resources from all sources and at all levels to finance sustainable forest management and provide adequate incentives to developing countries to advance such management , including for conservation and reforestation	{forest stewardship council} {rainforest alliance} {forest certification} {forest auditing}
Enhance global support for efforts to combat poaching and trafficking of protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities	{ecotourism} {community-based conservation} {community based conservation} {human-wildlife conflict}

Second update search: 97,814 document results

TITLE-ABS-

KEY ((terrestrial OR land OR inland OR freshwater) AND (biodivers* OR {species richness} OR bioeconom* OR bio-econom* OR {biological production} OR deforest* OR desertif* OR {earth system} OR {ecological resilience} OR ecosystem* OR eco-system* OR {trophic cascade} OR {trophic level} OR {trophic web} OR {threatened species} OR {endangered species} OR {extinction risk} OR {extinction risks} OR poach* OR {wildlife product} OR {wildlife products} OR {wildlife traffic} OR {wildlife market} OR {wildlife markets} OR {wildlife trafficking} OR {invasive species} OR {alien species} OR {land uses} OR {land use} OR {land uses} OR {land degradation} OR {soil degradation} OR {LULUCF} OR *forest* OR {land conservation} OR wetland* OR mountain* OR dryland* OR {mountainous cover} OR {protected area} OR {protected areas} OR {REDD} OR {forest management} OR {silviculture} OR {timber harvest} OR {illegal logging} OR {slash-and-burn} OR {fire-fallow cultivation} OR {tree cover} OR {soil restoration} OR {land restoration} OR {drought} OR {sustainable land

management} OR {mountain vegetation} OR {habitat restoration} OR {Red List species} OR {Red List Index} OR {extinction wave} OR {habitat fragmentation} OR {habitat loss} OR {Nagoya Protocol on Access to Genetic Resources} OR {genetic resources} OR {biological invasion} OR {biodiversity-inclusive} OR {forest stewardship council} OR {rainforest alliance} OR {forest certification} OR {forest auditing} OR {ecotourism} OR {community-based conservation} OR {community based conservation} OR {human-wildlife conflict})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 97,814 document results, compared to 94,819 document results for the original search.

The top 20 cited articles were reviewed. Five of the 20 articles do not closely reflect the theme of the SDG. These articles cover topics in relation to the field of medicine, climate change and human geography. The updated search query added three new articles to the top 20 most cited, these are all related to the theme of the SDG.

Top 20 most cited articles listed in Scopus from search string version 2:

**red text shows articles which do not closely match the theme of the SDG*

Title	Year	Source title	Cited by	EID
High-resolution global maps of 21st-century forest cover change	2013	Science	2870	2-s2.0-84887724460
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1292	2-s2.0-84871840877
Changes in the global value of ecosystem services	2014	Global Environmental Change	1257	2-s2.0-84901406973
Completion of the 2011 national land cover database for the conterminous United States – Representing a decade of land cover change information	2015	Photogrammetric Engineering and Remote Sensing	1050	2-s2.0-84929465562
Enhanced nitrogen deposition over China	2013	Nature	1001	2-s2.0-84874369007
Status and ecological effects of the world's largest carnivores	2014	Science	918	2-s2.0-84892185014
Defaunation in the Anthropocene	2014	Science	858	2-s2.0-84904795599
Communities and ecosystems: Linking the aboveground and belowground components	2013	Communities and Ecosystems: Linking the Aboveground and Belowground Components	797	2-s2.0-84890745455
The biodiversity of species and their rates of extinction, distribution, and protection	2014	Science	779	2-s2.0-84901642271
Comprehensive evaluation of antibiotics emission and fate in the river basins of China: Source analysis, multimedia modeling, and linkage to bacterial resistance	2015	Environmental Science and Technology	769	2-s2.0-84930632873
Climate change projections using the IPSL-CM5 Earth System Model: From CMIP3 to CMIP5	2013	Climate Dynamics	712	2-s2.0-84876880770

Global warming and changes in drought	2014	Nature Climate Change	699	2-s2.0-84890936362
Landsat-8: Science and product vision for terrestrial global change research	2014	Remote Sensing of Environment	687	2-s2.0-84896818071
Policy: Sustainable development goals for people and planet	2013	Nature	680	2-s2.0-84875346108
Paris Agreement climate proposals need a boost to keep warming well below 2 °c	2016	Nature	678	2-s2.0-84977119288
Global effects of land use on local terrestrial biodiversity	2015	Nature	676	2-s2.0-84926360683
Climate change impacts on global food security	2013	Science	669	2-s2.0-84881061285
Urban green space, public health, and environmental justice: The challenge of making cities 'just green enough'	2014	Landscape and Urban Planning	644	2-s2.0-84899128460
Nitrous oxide emissions from soils: How well do we understand the processes and their controls?	2013	Philosophical Transactions of the Royal Society B: Biological Sciences	633	2-s2.0-84878241957
Mercury as a global pollutant: Sources, pathways, and effects	2013	Environmental Science and Technology	624	2-s2.0-84878224971

In addition, a randomised sample of 20 articles was reviewed. Five of the 20 articles do not closely reflect the theme of the SDG. These five articles covered topics in relation to transport studies, climate change mitigation, marine ecosystems, polluted groundwater and (historical) studies on sediment layers.

Randomised sample of 20 articles listed in Scopus from search string version 2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Consequences of extensive habitat fragmentation in landscape-level patterns of genetic diversity and structure in the Mediterranean esparto grasshopper	2015	Evolutionary Applications	24	2-s2.0-84931956004
Mapping shallow lakes in a large South American floodplain: A frequency approach on multitemporal Landsat TM/ETM data	2014	Journal of Hydrology	24	2-s2.0-84896054289
Intensification of north american megadroughts through surface and dust aerosol forcing	2013	Journal of Climate	24	2-s2.0-84880685274
Constant diversification rates of endemic gastropods in ancient Lake Ohrid: Ecosystem resilience likely buffers environmental fluctuations	2015	Biogeosciences	18	2-s2.0-84949651050
Agriculture and biodiversity: A better balance benefits both	2016	AIMS Agriculture and Food	16	2-s2.0-85019363661
Integrating the system dynamic and cellular automata models to predict land use and land cover change	2016	International Journal of Applied Earth Observation and Geoinformation	11	2-s2.0-84997796226
Public Support for Policies to Reduce Risk After Hurricane Sandy	2014	Risk Analysis	10	2-s2.0-84903439272

Use of ecosystem information derived from forest thematic maps for spatial analysis of ecosystem services in northwestern Spain	2017	Landscape and Ecological Engineering	7	2-s2.0-84981187570
Predicting local colonization and extinction rates of freshwater mussels based on biological traits in a case of Lake Poyang	2017	Hupo Kexue/Journal of Lake Sciences	6	2-s2.0-85045299973
Marsh loss due to cumulative impacts of hurricane isaac and the DeepWater Horizon oil spill in Louisiana	2017	Remote Sensing	6	2-s2.0-85013659868
Mustard catch crop enhances denitrification in shallow groundwater beneath a spring barley field	2014	Chemosphere	5	2-s2.0-84896388712
Effect of arbuscular mycorrhiza on root self-repairing action of <i>Amorpha fruticosa</i> L. in coal mining subsidence land in arid areas	2014	Meitan Xuebao/Journal of the China Coal Society	4	2-s2.0-84908052662
Evaluating effects of everglades restoration on american crocodile populations in south florida using a spatially-explicit, stage-based population model	2014	Wetlands	4	2-s2.0-84905665250
Spatio-temporal variations of conservation hotspots based on ecosystem services in Xishuangbanna, Southwest China	2017	PLoS ONE	3	2-s2.0-85038430762
Assessing the effects of dam building on land degradation in central Iran with Landsat LST and LULC time series	2017	Environmental Monitoring and Assessment	2	2-s2.0-85010703228
Tracing organic matter sources in a tropical lagoon of the Caribbean Sea	2017	Continental Shelf Research	1	2-s2.0-85028871527
Improving identification of areas for ecological restoration for conservation by integrating USLE and MCDA in a gis-environment: A pilot study in a priority region northern Mexico	2017	ISPRS International Journal of Geo-Information	1	2-s2.0-85029283001
Unique clay assemblages in the Levant basin (Haifa Bay) subsurface reflect a land-sea transition zone and restricted environments throughout the last ~1 m.y.	2015	Marine Geology	1	2-s2.0-84945334099
Application of generalised regression neural networks in trip distribution modelling	2014	Road and Transport Research	1	2-s2.0-84918494260
Lake and watershed management: Issues and challenges in managing lake water quality	2016	Proceedings of the World Congress on New Technologies	0	2-s2.0-85045010540

Graphical summary of co-occurrence of terms from 2000 randomly selected results:

